

2 PLANT PHILOSOPHY AND PLANT SEX: ARISTOTLE TO ALBERTUS

What is plant sex? Although any botanist will have a ready answer to this question (male and female in plants are distinguished as bearers of gametes of different sizes), the question is, on closer inspection, particularly complex. The same ready answer serves as a definition of 'sex' in the animal kingdom too. More specifically, sex is defined in terms of anisogamy (reproduction via the fusion of two gametes of different sizes). As Lynn Margulis and Dorion Sagan explain:

the smaller of a pair of gametes is dubbed 'male'. Male sex cells are so designated, however, not only because of their smaller size but because of their propensity to move on their own.... the practical definition of *male* is simply an organism (gamont) that produces moving, relatively small gametes, often in profusion, but the term may be used for one of those gametes (or microgametes) itself. The term *female*, by general definition, usually refers to those gametes which stand still or those gamonts which produce sedentary gametes.¹

This scientific definition is obviously remote from the popular, everyday understanding of sex as the bare, biological fact of the difference between male and female. This scientific definition would have to be finessed to encompass all of those popularly identified as male and female (given, for example, that women who have achieved menopause, pre-pubescent boys and 'sterile' adults of all kinds do not, in fact, produce gametes). The popular definition refers mainly to types of embodied individuals, whereas the scientific definition bears specifically on the process of sexual reproduction (while acknowledging an ambiguity of sense). But whether we use Margulis and Sagan's definition or the popular one, neither in fact apply in any straightforward way to plants in the way that they apply to animals. Further, the investigation of the question of plant

sex forces us to ask questions about the meaning of ‘sex’ in general. It is not obvious what it means to say that plants are sexed, and one outcome of the further investigation of this problem is the realization that there are several different meanings of ‘sex’ in general and that their confusion has profound social and political implications.

The complexity of the question of plant sex is more or less systematically covered over in modern histories of botany, precisely where one may have expected to find it discussed: in the history of the theory of plant sexuality. As we indicated in the Introduction, traditional histories of the theory of plant ‘sexuality’, written from the standpoint of the present, take the idea of plant sexuality for granted as an empirical fact, search for early indications of its recognition and criticize the thinkers of past for failing to discover the knowledge of the present. A. W. Morton, for example, finds it strange that various otherwise important botanists (like Marcello Malpighi) failed to entertain the idea of plant sex.² For Morton, pollen is ‘manifestly the male element’³ in plant reproduction and he struggles to understand why botanists could not accept that straight away.⁴ In the most sustained treatment of the topic in English, Lincoln Taiz and Lee Taiz organize their book *Flora Unveiled* (2017) around two ‘basic questions’: ‘why did it take so long to discover sex in plants, and why, after its proposal and experimental confirmation in the late seventeenth century, did the debate continue for another 150 years?’⁵ Like Morton, they wonder why it took so long for the past to catch up with the present, surprised that neither Aristotle nor his contemporary Theophrastus, ‘despite their considerable resources’, advanced the problem of sex in plants: ‘The failure of these two luminaries to recognize the more general role of pollination in hermaphroditic flowers has long puzzled historians of botany.’⁶

We are mainly speaking here of the Western botanical tradition. This tradition, in its developed theoretical aspects up until the scientific advances of the eighteenth century, is fundamentally Aristotelian. Traditional histories of the theory of plant sexuality understand the progress towards a properly scientific botany as dependent on its liberation from the philosophical commitments of the early, Aristotelian plant philosophers (especially Nicolaus of Damascus, Albertus Magnus and Cesalpino) and in terms of the rejection of analogies with animals in favour of a literal approach to plant sex and reproduction. Plant science extricates itself from plant philosophy and at last plant sexuality becomes obvious. In effect, according to the traditional histories, ‘plant philosophy’ was the problem all along.

But approaching the history of the theory of plant sexuality from a positive plant philosophy perspective, in which the meaning of ‘sexuality’ is not taken for granted, things look rather different. We see that the zoocentric model in the study and understanding of plant life is nowhere more obvious than in discussions of plant sex, which are indeed archetypes of the zoocentric model. We see the extent to which that zoocentrism is the outcome of a history of explicitly analogical reasoning in the study and philosophy of plant life – a history of analogies between animals and plants, with animals supplying the ‘source’ terms with which plants are compared. But we also see that the question of plant sex (or the question: Do plants have male and female?) cannot be addressed without first asking a more fundamental, philosophical question: what are ‘male’ and ‘female’? We see that plant sex was never ‘obvious’ for the plant philosophy tradition precisely because it paid attention to this prior, philosophical question. We see how the most important moments in the ‘discovery’ of plant sexuality are connected to this philosophical question and how the plant philosophers concerned reflected explicitly and critically on the analogy with animals, rather than taking it for granted. In plant science the zoocentric and anthropological terms in which plant sex is discussed now almost always get a free pass, and the conflation of the different senses of ‘sex’ and ‘male’ and ‘female’ sows confusion in discussions of plant reproduction and ecology. This same conflation in popular discourses sows political confusion and leads to the reactionary biologization of social phenomena.

The emergence of the strikingly original works by Marder and Coccia, especially, in the early twenty-first century could give the impression that plant philosophy is a very recent phenomenon, even where it revives ancient ideas (such as Marder’s treatment of the idea of the vegetative soul). But in this chapter we will reconnect it with a much longer tradition, drawing out one particular thread – discussions of plant sex – in order to raise the philosophical questions about ‘sex’ and ‘male’ and ‘female’ again in the context of contemporary plant philosophy and science. If, as we have said, discussions of plant sex are archetypes of the zoocentric model, we need to understand the analogical forms that underlie this, and the historical basis of explicitly analogical reasoning is to be found in Aristotle’s philosophy. In this chapter, then, we will look at the different kinds of analogy in Aristotle’s philosophy generally, at the analogical models employed in Aristotle’s zoology specifically and at the idea of the vegetative soul. It is this idea of the vegetative soul, we will argue, that makes analogies between animals and plants possible and

which is (as we shall see in the next chapter), contrary to the traditional histories of botany, one of the bases of the developments in the theory of plant sexuality in the eighteenth century. We will identify the specificity of the analogical form in questions concerning male and female in plants as Aristotle poses them, and in the treatment of the same question in the Aristotelian botanical tradition (or plant philosophy) represented by Theophrastus, Nicolaus of Damascus and Albertus Magnus. Perhaps this seems remote from contemporary social and political questions about sex, but it is not. Because it is, perhaps surprisingly, in Aristotelian plant philosophy that we find the reflection on the different meanings of sex that we might do well to reintroduce into contemporary debates.

Forms of analogy and the maintenance of difference

If botany, from its classical sources and well into the eighteenth century, was often based on a series of analogies with animals, this is no doubt in part (if not wholly) due to the overwhelming importance of Aristotelian philosophy in theoretical botany.⁷ Aristotle employed analogies throughout his philosophical works and especially frequently in his zoology. In the programmatic early remarks in both *History of Animals* and *Parts of Animals* it is made clear that analogical comparison is a formal methodological principle in Aristotle's zoological investigations. In both works it is stated that within 'genera' (by which he means any group constituted by what its members have in common; this is not the modern taxonomic concept of genus) things differ from each other by degree. Aristotle calls these differences of the 'more and the less', or of 'excess and defect', whereas across genera things differ in kind and any sameness between them is only 'in the way of analogy':

For all kinds [*genōn*] that differ by degree and by *the more and the less* have been linked under one kind [*heni genei*], while all that are analogous have been separated. I mean, for example, that bird differs from bird by *the more* or by degree (one is long-feathered, another is short-feathered), but fishes differ from bird by analogy (what is feather in one is scale in the other). But to do this for all is not easy, since the similarity in most animals is by analogy.⁸

Analogies pick out specific aspects of animal bodies where there is a commonality of function ('parts' being explicitly correlated with functions⁹):

By 'analogously' I mean that some have lungs while others have not lungs but something else instead which is to them what lungs are to the former; and some have blood while others have the analogous part that possesses the same capability as blood does for the blooded.¹⁰

The concept of analogy in modern biology (similarity of function in parts of unrelated genera) is derived with surprisingly little change from this. We can call this a 'formal' analogy, or an analogy of relation (of part[s] to whole) in the different species – in the logical sense – being compared.¹¹

Aristotle sometimes points out such analogies where they are relatively obvious, for example: 'The quadruped vivipara instead of arms have forelegs. This is true of all quadrupeds, but such of them as have toes have, practically speaking, organs analogous to hands; at all events, they use these forelimbs for many purposes as hands – except for the elephant.'¹² He also uses analogy, implicitly and explicitly, where the relation between parts of different genera is not so obvious, but where there *must be* analogies as all animals must have the parts that correspond to the functions that define them as animals.¹³

Aristotle also uses what we can call 'substantive' analogy, which works via the identification of shared properties, implying a shared nature. This can be the basis for the definition or identification of a genus.¹⁴ Mary Hesse cites Aristotle's example, in *History of Animals*, of the analogy between (human) bone and fish bone or spine, suggesting that the analogy implies that they share an 'osseous' nature (for which there is no specific word).¹⁵ Finally, Hesse identifies a third, 'metaphysical' kind of analogy. In *De Anima*, for example, Aristotle explains a distinction between two sorts of actuality, the example being the difference between the actuality of knowledge and the actuality of contemplating: 'as waking is analogous to contemplation [both actualities], sleeping is analogous to having knowledge without exercising it [both potentialities]'.¹⁶ In the *Metaphysics*, the same philosophical problem – the definition of actuality – is also explained with recourse to analogy, but in a different way. Contrasting actuality with potentiality explicitly, and drawing a series of actual-potential antitheses (e.g. building: capable of building; waking: sleeping; seeing: eyes shut but capable of sight; what is shaped out of matter: the matter), Aristotle says: 'Let actuality be defined by one

member of this antithesis, and the potential by the other. But all things are not said in the *same sense* to exist actually, but only by analogy... for some are as movement to potentiality, and the others as substance to some sort of matter.' This is an example of a case in which we must not seek a definition (here, of actuality) 'but be content to grasp the analogy'.¹⁷ The *Physics* supplies us with another such example, concerning 'underlying reality' [*hupokeimenē phusis*].¹⁸ Hesse connects this to the earlier discussion of 'one' in the *Metaphysics*, where Aristotle writes that

some things are one in number, others in species, others in genus, others by analogy; in number those whose matter is one, in species those whose formula is one, in genus those to which the same figure of predication applies, by analogy those which are related as a third thing is to a fourth... things that are one by analogy are not all one in genus.¹⁹

In these examples, as Hesse says, analogy is the basis of a kind of 'definition' where definition *per genus et differentiam* is impossible, because what is being defined is not itself either genus or species, or more specifically because there is no higher genus ('actuality', 'one', 'underlying reality', and perhaps all metaphysical terms are not genera). It also applies, as here, when the reduction to univocity is impossible (the term that is 'defined' does not apply to the examples used to explain it in the same way, precisely because the term crosses genera), but where the term is also not absolutely equivocal (having different meanings).²⁰ This may seem to be far from our topic of vegetal sex, but, as we shall see, 'male' and 'female' in plants are in fact 'defined' in this way in the Aristotelian botanical tradition.

The peculiarity of analogy in all of its forms (formal, substantive and metaphysical) lies in the first instance in its being the method to compare and make demonstrations across genera (in the sense identified earlier: any group constituted by what its members have in common).²¹ One important consequence of this, and what is crucial to its legitimate use, is that the terms of an analogy (the things compared) are not rendered *identical* by the analogy; their difference is maintained, despite the postulation of a similarity.

So what kind of analogy is at work in analogies between animals and plants? And what possible basis is there for such analogies in the first place?

The vegetable soul

Aristotle's conception of the hierarchically arranged unity of the natural world assigned vegetable life a place that both connected and separated it from animal and human life. One paragraph from Chapter VIII of the *History of Animals* sets this out clearly:

Nature proceeds little by little from things lifeless to animal life [*ek tōn apsuchōn eis ta zōia*] in such a way that it is impossible to determine the exact line of demarcation, nor on which side thereof an intermediate form should lie. Thus, next after lifeless things comes the plant [*tōn phutōn*], and of plants one will differ from another as to its amount of apparent vitality; and, in a word, the whole genus of plants, whilst it is devoid of life as compared with an animal, is endowed with life as compared with other corporeal entities. Indeed, as we just remarked, there is observed in plants a continuous scale of ascent towards the animal. So, in the sea, there are certain objects concerning which one would be at a loss to determine whether they be animal or vegetable [*ē phuton*].²²

This 'continuous scale of ascent', although it may be traced through physical features, is secured philosophically, for Aristotle, through the postulation of the three 'parts' of the soul (*psuchē*) or three kinds of soul: nutritive (or vegetable), perceptive (or sensitive) and rational. Aristotle postulates that soul is a substance as 'the form of a natural body which has life in potentiality'. By 'life', he says, he means 'that which has through itself nourishment, growth, and decay'.²³ A natural body that has life in potentiality is one which is 'organic', which means both having organs and, as is later clarified, that the body is an organ of the soul.²⁴ In the next chapter, living things are defined as those that have at least one of the following: 'reason, perception, motion and rest with respect to place, and further the motion in relation to nourishment, decay and growth'. As plants have the latter 'motion', 'even plants, all of them, seem to be alive'.²⁵ The part of the soul corresponding to this motion is the 'nutritive capacity' (*threptikon dunatai*). This also came to be known, via the Latin, as the 'vegetative' part of the soul. This nutritive capacity is the principle through which all living things are alive. All animals, in addition, have perception or sensation (which implies also desire, appetite, spirit,

pleasure and pain);²⁶ some (i.e. human) animals, as well as the nutritive and perceptual faculty, also have the faculty of understanding and reason (*to dianoētikōn te kai noûs*).²⁷ Thus the ‘nutritive soul’ (*threptikē psuchē*) is ‘the first and most common capacity of the soul in virtue of which living belongs to all living things, a capacity whose functions are generating and making use of nutrition.’²⁸

There is a kind of substantive analogy at work here, to the extent that the comparison of nutrition and generation in plants and animals implies a shared nutritive nature giving rise to the definition of the vegetative soul. And this shared organic, natural being and the shared principle of nutrition, growth and generation are then itself the basis for any formal analogy between plants and animals. This is evident in Aristotle’s first such analogy, which is also the first mention of plants in *De Anima*: ‘And even the parts of plants are organs, although altogether simple ones. For example, the leaf is a shelter of the outer covering, and the outer covering of the fruit; and the roots are analogous to the mouth, since both draw in nourishment.’²⁹ The shared nutritive capacity (or shared nutritive soul) is the basis of the presumption that there must be something in plants analogous to the parts of animals that serve the nutritive function.

The basic formal analogy between mouth (and head) and roots is made several times, but the inexactness of the analogy is shown by its variants and by the fact that anything more than the vaguest comparison of the nutritive parts is quickly strained.³⁰ Indeed, because plants have fewer functions and correspondingly fewer organs than animals (fewer heterogeneous parts), analogies between them are inevitably very limited and Aristotle concludes that ‘the configuration of plants is a matter then for separate consideration.’³¹ Even here, then, in what is arguably the beginning of the zoocentric tradition in discussions of plants, the inadequacy of animal models and the specificity of plant forms is already implicitly admitted.

What are ‘male’ and ‘female’?

In principle, the shared vegetative soul in animals and plants, associated with the functions of nutrition, growth and generation, means that, for Aristotle, there must also be analogies between plant and animal *generation*. In *History of Animals* Aristotle notes that a common property of plants and animals is that in both groups some are generated from seed/parent and

some are spontaneously generated.³² In *Generation of Animals* an analogy is drawn between animals' production of young and plants' production of seed and fruit, as if the seed were the offspring, but more often the analogy is between plant seed and semen, and Aristotle admits that 'There is considerable difficulty in understanding how the plant is formed out of the seed or any animal out of semen [*tou spermatos tōn phutōn ē tōn zōiōn hotioun*].'³³ He also compares animals' eggs to the seeds of plants, and in so doing broaches the difficult issue of the applicability of the male/female division to plants. However, in this instance the form of the analogy is significantly different from almost all of Aristotle's other zoological analogies. As we have noted, Aristotle's zoological analogies (between animals) are primarily formal analogies between functionally similar parts. Such analogies between functionally similar parts of animals and plants are from the beginning more difficult in the case of male and female because of the lack of any obvious similarity between any parts, and the complete absence, as far as Aristotle is concerned, of any specifically male and female parts in plants. This means that the analogy between 'male' and 'female' in plants and animals will of necessity take a completely different form.

Some modern historians of botany find it hard to understand why Aristotle did not recognize the 'sexuality' of plants.³⁴ But when we understand what, for Aristotle, 'male' and 'female' are and what roles they play in reproduction, we can see that the idea of male and female plants posed a particular problem for him. In *Generation of Animals* Aristotle's aim is to explain both the specific contributions of male and female to generation and the generation of male and female animals. In this context the male and the female are defined, primarily, in terms of difference in capacity, role or function:

Male is what we call an animal that generates into another, female that which generates into itself.... Male and female differ by definition [*logon*] in having different capabilities [*dunasthai*], and by appearance in certain parts. They differ by definition in that the male is that which can generate into another... while the female is that which generates into itself and out of which the generated offspring is produced while present in the generator. Since they are distinguished by capability [*dunamei*] and by particular function [*ergōi*], and since there is need of organs for every performance of function, and the organs for the capabilities are the parts of the body, there must be parts both for producing young and for coupling, and these parts must be different

from each other. For although male and female are predicated of the animal as a whole, nevertheless it is not male or female in respect of all of itself but in respect of a particular capability and a particular part.³⁵

As is well known, Aristotle claims that the female contributes the matter, while the male imparts the motion and the soul in generation. And this is another way, indeed, of *defining* the male and the female: ‘the power of making the sensitive soul’ is the essence of what being male is – it is what it is to be a male.³⁶ Aristotle makes this latter claim in the context of the problem of why animals need males to reproduce, given that females possess the same soul as males and provide the material for the embryo:

The reason is that the animal differs from the plant by having sense-perception [*aisthēsei*]; if the sensitive soul [*aisthētikēs psuchēs*] is not present, either actually or potentially, and either with or without qualification, it is impossible for face, hand, flesh, or any other part to exist; it will be no better than a corpse or a part of the corpse. Thus if it is the male that has the power of making the sensitive soul, then where male and female are separate it is impossible for the female to generate an animal from itself alone, for the process in question was what being male is.³⁷

If ‘being a male’ just is the power of making the sensitive or perceptive soul (in English translations ‘sensation’ and ‘perception’ both translate *aisthēsis*) there cannot by definition be any male plants, because no plants have such a power. Although Aristotle must admit that plants have parts of some kind (as we shall see, what their parts are is a very difficult question for him and his contemporary Theophrastus), there can be for him no ‘sensitive’ parts, which we must assume to include the sexual parts.³⁸

Together, Aristotle’s definition of an animal and his definition of a male seem to contradict his claim that some animals, however, have no male and female.³⁹ How can this be the case? As Aristotle himself asks, what in these cases corresponds to the female material principle that *Generation of Animals* takes to be located in the residue secreted by female animals, and ‘what is the moving principle which corresponds to the male?’⁴⁰ There are in fact two problems here. First, how are such animals still animals, if they do not have the male that imparts the sensitive soul? Some remarks suggest that such ‘animals’ might represent the point of indifference between animal and plant, for example the

testacea: ‘because their being is almost like plants there is no male and female any more than in plants.’⁴¹ Some testacea, according to Aristotle, are formed spontaneously while ‘some emit a sort of generative substance from themselves’, but this is not ‘real semen [*sperma*]’. To understand the generation of such creatures we need to ‘grasp the different methods of generation in plants; some of these are produced from seed, some from slips, planted out, some by budding off alongside’; testacea ‘participate in the resemblance to plants’ in this manner.⁴²

However, as all living things are defined as those that have ‘motion [*kinēsis*] in relation to nourishment, decay and growth’ and the male is said to be that which imparts movement (more generally; not specifically the sensitive soul) to the female material principle,⁴³ there can be no living thing at all – animal or plant – that does not have the male principle. It seems, then, that what Aristotle means when he says that some animals have no male and female is that in some animals there is no *separation* between male and female as ‘male and female are distinct in the most perfect [animals], and... these [male and female] are first principles [*dunameis archas*] of all living things whether animals or plants, only in some of them these principles are separated and in others not.’⁴⁴ More specifically, all plants and some animals contain the male principle within themselves mixed with the female. Aristotle sometimes seems to suggest that in such cases (e.g. with bees and certain fishes) it may appear that the female generates without the male, but in fact the ‘female’ contains the male principle within herself ‘as plants do’ and this is sufficient, without the contribution of a *separate* male, for ‘fotation’ to occur with the relation between nourishment and heat.⁴⁵ In sum:

In all animals which move about, male and female are separate; one animal is male and another female, though they are identical in species, just as men and women are both human beings, and stallion and mare are both horses. In plants, however, these faculties [*dunameis*] are mingled together; the female is not separate from the male; and that is why they generate out of themselves, and produce not semen [*gonēn*] but a fotation [*kuēma*] – what we call their ‘seeds’.⁴⁶

Whatever these male and female ‘faculties’ or ‘principles’ are, they are obviously not identifiable with male and female *individuals* or male and female *parts*, not least because they are said to be the cause of male and female individuals and male and female parts.⁴⁷ This suggests that there are

several senses of ‘male’ and ‘female’ in Aristotle’s remarks on living beings, and that ‘male’ and ‘female’ are defined according to what Hesse calls a metaphysical analogy, as they are neither univocal terms (‘male’ and ‘female’ are not always meant in the same sense) nor absolutely equivocal. ‘Male’ and ‘female’ are (i) *metaphysical principles* of the generation of life which may be found mixed together in one plant or animal or found in separate individuals, (ii) separate kinds of *individuals* in some kinds of animals and (iii) separate kinds of *parts* of individuals in some kinds of animals. When Aristotle says that there is no male or female in plants he means no male or female in senses (ii) and (iii). Aristotle concedes that individual plants are sometimes *called* ‘male’ and ‘female’ in sense (ii) – a particular kind of plant may be said to divided into ‘male’ and ‘female’ individuals – but only by a kind of analogy. In *Generation of Animals*, for example, he says that, although there is no male and female in plants and some animals (e.g. testacea),

they have come to be called male and female in virtue of resemblance and analogy, for they [animals like testacea] have a small differentiation of this sort. Among the plants too there are in the same kind of tree some that are fruit-bearing and some that do not bear themselves but contribute to the concocting of others’ fruit, as occurs with the fig and caprifig.⁴⁸

But Aristotle does not ever suggest any analogy of male and female *parts* (sense iii) or of functions of parts – that is, no analogy of the form that is otherwise common and indeed methodologically central in his zoology, and (as we shall see in later chapters) is the form that dominates the discussion and understanding of male and female in plants from the seventeenth century to the present day.

However, Aristotle’s hylomorphism (the metaphysical doctrine that all beings are composed through two principles, form and matter) and the speculative metaphysics of generation based on it require him to think male and female principles, as principles of life, in sense (i), even in plants. Every plant of every kind, just because it is a living thing, must be understood to contain within in it – ostensibly in reality, not analogically – male and female principles. But what does this mean – ‘male’ and ‘female’ *principles*? Discussion of the male and female principles in the literature on Aristotle tends to approach the question with the idea of (separate) male and female animals (sexed individuals) in mind, most often human animals.⁴⁹ But the problem is really to identify and explain

what male and female principles are across all of the animal kingdom, including where they are not separate, *and in plants*. Indeed, the need to answer the question in relation to living beings where (according to Aristotle) ‘male’ and ‘female’ are not separate and there are no ‘male’ and ‘female’ parts means that the question ‘do plants have male and female?’ becomes, of necessity, a more fundamental question about the nature of the metaphysical principles of male and female themselves.

As we shall see below, this becomes more explicit in the later Aristotelian texts that deal with the question of male and female in plants, especially those of Nicolaus of Damascus and Albertus Magnus. But let us just note, for now, the result of the plant-philosophical orientation of our discussion of Aristotle. The analogies between animals and plants in Aristotle’s work have been shown to be mostly of the formal type (analogies of part and function across genera), and these have been shown to rest on a kind of substantive analogy of capacities that gives rise to the definition of the vegetative soul. As Aristotle understands the functions of nutrition, growth and generation – precisely, the functions of any living organism – in terms of organs or parts, his analogies try to identify the relevant organs and parts of plants, even as he recognizes that plants do not, in fact, have ‘parts’ in the way that animals do. This general point is nowhere more clearly articulated than in the attempt to find analogies in the generation of animals and of plants. Indeed, this of necessity leads Aristotle away from analogies of parts to the more fundamental, metaphysical question: what are ‘male’ and ‘female’ such that these principles can be said to be part of vegetable life? As the answer requires us to distinguish between different senses of ‘male’ and ‘female’ these terms are shown to be defined according to what Hesse calls ‘metaphysical analogy’. Like all metaphysical terms they are neither univocal nor absolutely equivocal, and the philosophical challenge is to maintain their different senses while understanding the nature of their generality. Thus, the discussion of what will later be called plant ‘sex’ turns out to be of greater philosophical interest than its restriction to botany might have us believe.

What we call ‘male’ and ‘female’

In the surviving texts and fragments that comprise the Aristotelian tradition in botanical philosophy the question of male and female in plants is considerably developed – indeed, it becomes one of the central

questions. It is interesting, however, that Aristotle's contemporary Theophrastus, whose two books on plants (*Enquiry into Plants* and *Causes of Plants*) echo the form and procedure of Aristotle's zoology in many ways, studiously avoids all question of the metaphysical principles of male and female. *Enquiry into Plants* begins (like *History of Animals* and *Parts of Animals*) with methodological considerations, and the extent of the validity of plant-animal analogies is at the heart of these: 'In considering the distinctive characters of plants and their nature generally one must take into account their parts, their qualities, the ways in which their life originates, and the course which it follows in each case: (conduct and activities we do not find in them, as we do in animals).'⁵⁰ But we are immediately faced with the problem that it not obvious what should count as a 'part' of a plant. As something belonging to a plant's characteristic nature it seems that a 'part' should be something that, once it has appeared, is permanent – like the parts of animals. But many of the parts of plants, and especially those concerned with reproduction, are ephemeral. If we count these the number of parts will be indeterminate and constantly changing; if we do not count them essential features of the plants will be left out.⁵¹

Theophrastus's answer to this stresses the need to be cautious with any plant-animal analogy while also suggesting that it will often be most fruitful to proceed precisely by such comparisons. In some cases, where there exist no special names for particular parts of plants, resemblance (of part and of function) dictates that the names of corresponding parts of animals are borrowed – for example 'fibre' and 'vein'.⁵² This and other analogies with animals are justified because, in general, animals are simply better known, and it is by the better known that we must pursue the unknown. Analogies concerning generation are identified as particularly challenging:

But perhaps we should not expect to find in plants a complete correspondence with animals in regard to those things which concern reproduction any more than in other respects; and so we should reckon as 'parts' even those things to which the plant gives birth [*ta gennōmena*], for instance their fruits, although we do not so reckon the unborn young of animals.⁵³

Does Theophrastus pursue an analogy between male and female in animals and plants? Certainly, he speaks relatively often of male and

female plants (principally trees, his model for all plants), but it is striking that this is not part of any functional analogy. Rather, ‘male’ and ‘female’ are most often for Theophrastus folk-designations for different types of plants of what we would call the same species. As such, he often notes that what counts as ‘male’ and what counts as ‘female’ is differently determined in different contexts and that these multiple determinations are sometimes incompatible. Some say, for example, that in plants of the same kind the male flowers while the female does not.⁵⁴ Some say males do not bear fruit but others say that only males do.⁵⁵ But however they are identified, the difference between ‘male’ and ‘female’ is, according to Theophrastus, ‘of the same character as that which distinguishes the cultivated from the wild tree, while other differences distinguish different forms [*eidōs*] of the same kind [*homogenōn*],’⁵⁶ and the difference between fruitless and fruit-bearing, flowering and flowerless (which can also be markers of ‘male’ and ‘female’) is said to be due to the position of the plant and the climate of the district.⁵⁷

In Book II of *Enquiry into Plants*, when Theophrastus considers propagation (spontaneous growth, growth from seed, root, a piece torn off, from a branch or a twig, from the trunk and so on), there is no attempt to make any functional analogy between male and female animals and male and female plants.⁵⁸ The only exception – and the most frequently cited example of male and female trees in the ancient literature – is date palms: ‘With dates it is helpful to bring the male to the female; for it is the male which causes the fruit to persist and ripen.’ He describes the process of shaking the ‘dust’ of the male flower over the female fruit, allowing the male to ‘render aid’ to the female – ‘for the fruit bearing tree is called “female”’.⁵⁹ But even here Theophrastus mentions no possible analogy with animal generation, or with the nature of the distinction between male and female animals.

In Book III, the differences between so-called male and female trees in general are said to concern the qualities of the timber. According to the woodmen any wood of a male tree cut with an axe ‘gives shorter lengths, is more twisted, harder to work, and darker in colour’, while the female gives better lengths and contains the heart wood that is less resinous and soaked with pitch so smoother and of straighter grain.⁶⁰ In various trees the wood of the female is said to be whiter, softer and easier to work; the male thicker, harder, unbending and inferior in appearance.⁶¹ Thus if there is an analogy between male and female in the animal and the plant worlds, it is not one based on any attempt to find analogous functions in

generation. Rather, it is an analogy based on the identification of common properties assumed to be characteristic of male and female animals, and perhaps male and female humans above all.⁶²

In *Causes of Plants*, devoted to generation and propagation, we again find no functional analogy between male and female animals and plants (even though Theophrastus speaks more than once of ‘pregnant’ trees).⁶³ The difference between male and female is once again understood to be qualitative, and Theophrastus is careful to play down the analogy with animals. Perhaps the fact that wild trees produce more seed is explicable in terms of their

being as it were male [*hōsper arrena onta*] and in their nature closer in texture and drier, for cultivating and good feeding have an effeminating effect [*diathēlounousi gar hai katergasiai kai hai trophai*]. This remark however, is to be taken as resting on a certain resemblance, and the parallel is pretty remote [*touto men ouv hōs kath’ homoiotēta tina legesthō, porrōter ō keimenon*].⁶⁴

If female trees are more fruitful than males they are nevertheless not so hot, ‘as one can assume from their similarity to the female in animals, even if here the word has a different sense.’ The relation between the male and the female date palm is said to be ‘something similar’ to the relationship between fish, ‘when the male sprinkles his milt on the eggs as they are laid. But resemblances can be found even in things widely separate.’⁶⁵ In sum, then, Theophrastus treats male and female as popular names for types of plants (principally trees), and his descriptions suggest that the popular designations are based on an analogy between some of the imagined properties of male and female animals, but not on their roles in reproduction.⁶⁶ The metaphysical principles of male and female play no role in Theophrastus’s discussion.

However, the subsequent, and more overtly philosophical tradition in botany foregrounds the metaphysical aspect. The first-century BCE *De Plantis* of Nicolaus Damascenus is thought to be a re-elaboration of Aristotle’s lost work of the same title and was until relatively recently misattributed to Aristotle.⁶⁷ Only fragments of translations of the lost Syriac translation of Nicolaus’s *De Plantis* survive, but the various translators who also commented on the texts elaborate on it for us. These texts seem to say contradictory things about the possibility of male and female in plants, which they acknowledge is an important question (along

with the questions of whether plants have souls and/or sensation, whether they breathe and whether they sleep). All of the different translations frame the question as a response to a claim attributed to Empedocles (translated here from Bar Hebraeus) that plants ‘are males and females at the same time, because in each of them these two natures are mingled and united.’⁶⁸ Ishāq ibn Ḥunayn’s text says (in translation, at least), ‘I wish I knew whether plants... have males and females, or something in which male and female are combined’,⁶⁹ suggesting two options. Some note the objection that the latter seems to presuppose the former, because that which is mingled must first have been separate.⁷⁰ Most then suggest that the answer to this question lies somehow in the comparison (which Ibn At-Tayyib attributes to Aristotle⁷¹) of birds’ eggs and seeds. Bar Hebraeus’s commentary, in his *Butyrum Sapientiae* (*The Cream of Wisdom*), is most interesting on this point:

If a reader calls a body male that by nature has a capacity for moving a matter to receive a form similar to its own, and female a body that has a passive capacity for receiving a form in the above mentioned way, then plants have male and female, and it is possible that a single plant is male and female *at the same time*: female because the said matter is generated in it, and male because the said moving capacity originates in it.

But if it is not so, but the reader calls a body male from which a matter exudes by means of organs intended for that, which acts upon a matter present in the recipient, and he *calls* the recipient female, then there is neither male nor female in plants.

Let us agree, then to call female the capacity that secretes a residue from the plant that enters into the constitution of the like generated out of it, and male the forming capacity that shapes the residue.⁷²

In animals, he continues, these two capacities are united in the foetus after the intercourse of two separate individuals. Similarly, in every egg ‘there is a generative capacity and a capacity for receiving generation; and therefore a chick is generated in each of them. The process is similar to that of seeds in plants.’⁷³ Bar Hebraeus seems, in this way, to make a distinction between, on the one hand, male and female as separate individuals or kinds, one (male) exuding a matter into the other, the recipient (female); and, on the other hand, male and female as capacities (the capacity to give and receive form respectively) which may both be

inherent in one individual – a distinction that would resolve the apparent contradictions in *De Plantis*. Plants do not have male and female in the first sense, but they do in the second.

Hunayn's text similarly denies that plants have separate male and female in the first sense, as 'simple components' that become mingled, while affirming that they do have male and female in the second sense:

But as regards the mingling of males and female in plants, we must understand it differently, for the seed of a plant resembles a fetation, and that is a mingling of male and female. And just as in an egg there exists the power to generate a chicken, and *along with it* the material to feed it up to the time of its completion and its hatching, while *on the other hand* the female lies [sic] one egg at a time, so *it is with the seed* of a plant.⁷⁴

These different senses seem to correspond to the distinction in Aristotle's works between male and female as separate kinds of individuals and male and female as metaphysical principles. In animals the metaphysical principles reside in the different individuals, to become mixed in intercourse and its products (the foetus and the egg). In plants the metaphysical principles are already found mixed in the plant, or more precisely in the seed, which in this respect resembles an egg.⁷⁵ In Bar Hebraeus's commentary these metaphysical principles of male and female are based on the more fundamental distinction between activity and passivity, making explicit what we have argued is implicit in Aristotle's texts.

This is also the case with Albertus Magnus's thirteenth-century discussion of the problem of the sex of plants in his *De Vegetabilibus*. Albertus explicitly claims to resolve the contradictions on this question that arise from a survey of the ancients, principally the contradiction between Aristotle (who, according to Albertus, says there is no male and female in plants) and Theophrastus and others (who identify various male and female plants) or say that male and female are mixed in plants (a position that logically requires them to have been separate before they are mixed). Albertus effectively identifies the cause of these contradictions in the possibility of different ways of speaking about male and female in relation to plants and animals.

If, he writes, we define 'male' as that which from its own seed generates in another, and 'female' as that which takes seed from another and

generates in itself, there can be no doubt but that 'sex is not found in plants.'⁷⁶ Thus Albertus affirms Aristotle's definitions of male and female in animals and the consequence of them for the possibility of sex in plants. But is it possible, he continues, that some shared sexual properties or characteristics (*aliquas proprietates sexum participantium*) can be found in plants? Albertus then lists some of the qualities that Theophrastus had used to describe male and female plants (though he does not name Theophrastus), with the addition of an Aristotelian explanation for these different qualities:

So, the masculine, because of its forming and stamping [role] is harder and drier and as a consequence rougher to touch. The feminine however, because it is formed and it is stamped, has properties the opposite of these, so softness, moistness, smoothness, since those properties do well to bear the act of forming. And we find in plants that are called masculine that every one that is reproduced from these is stronger and rougher. And every one that is generated from those which are called female is softer and smoother.

His conclusion – it is thus clear that 'the sexes may not be distinct in plants'⁷⁷ – may be contested by the reader confused by the transfer of the definitions of male and female (forming and stamping/formed and stamped) into plants, but it is only the qualities that are *associated with* the roles of male and female that are transferred to those plants called male and female, not the roles themselves.⁷⁸

A few chapters later, Albertus returns to those ancients who say that there are male plants and female plants (e.g. olives and peonies). These authorities assert that the generation of every living thing involves a power proper to that thing (*agens proprium*), and that the active power cannot be the same as the passive 'power' because that would involve the identity of opposites (principally, action would be the same as passion). According to this view (which Albertus associates with Plato), 'In plants therefore active generation and passive generation in a single species will be separated by substance.'⁷⁹ Albertus counters this view with Aristotle's claim that male and female are inseparable accidents peculiar to animals, and not plants,⁸⁰ and his claim, in his book on animals (presumably *Generation of Animals*), that the distinction of the male sex from the female sex is only to be found where there is a sensitive soul. If it was only the vegetative soul that was responsible for generation (in animals,

presumably) no distinction of sex would be necessary.⁸¹ The power responsible for the sensitive soul cannot be in the passive ‘seed’ of the female, as the phenomenon of wind eggs shows, so in the more perfect generation of animals (more perfect than plants, that is), something in addition to the female seed is required to contribute the sensitive soul, whereas (it is implied) this is not necessary in the plant. Thus, Albertus says, it is Aristotle’s argument that plants do not combine (*conveniat*) male and female in generation, and if the contrary position is said to be attributed to him, it can only be that he means it metaphorically.⁸²

This metaphorical sense is further explained in the next paragraph. There is a true sex or true power or virtue (*virtus*) of sex and an imitation of this virtue or power. The true sex (one sex ‘forming and creating’ the sensitive soul in the animal seed, the other ‘worked and formed’ in the organ of its soul) is ‘by no means in plants’. The action of the sun on the seed of a plant is analogous (*similitudo*) to the action of the male power of sex on the passive female seed, but this is the only sense in which there can be said to be sex in plants, ‘not a true version, but a distant imitation’. This is reiterated at the end of the chapter: ‘It is therefore the conclusion of this question, according to the Aristotelian tradition, that sex in no way combines [*convenit*] in plants, in the same way as it combines in animals’ but there is a ‘remote analogy’ (*similitudinem remotam*), and this distinction between true sex and its remote analogy resolves the contradiction in the disagreements of the ancients on this question. In whatever way the ancients conceived of the distinction of sex in animals, it is only by analogy that this distinction can be said to apply to plants.⁸³ Albertus thus goes further than Nicolaus in denying that plants have male and female, because he denies not only that they are male and female *individuals* or substances (separate or mixed) in any real sense but also that the metaphysical principles (or ‘virtues’) of male and female can be said to belong to plants even analogically.

Once again, we see that the question of male and female in plants cannot be addressed by these authors without tackling the more fundamental question about the nature of the metaphysical principles of ‘male’ and ‘female’ themselves. Or, to put this in another way, the very difficult question of what we mean by ‘male’ and ‘female’ across genera, across different forms of life, is forced to philosophical attention by the study (albeit it is here exclusively theoretical) of plants. It is not that the meaning of ‘male’ and ‘female’ is clear, and that the problem is only to discover whether there are male and female plants. It is not, then, an

empirical problem. Rather, the life of plants complicates the question of what is meant by 'male' and 'female', giving rise to a philosophical problem. This, in a nutshell, is the central thesis of this book.

When we focus on this philosophical question, it is hard not to conclude that the positing of male and female *principles* in all living things is itself the result of an implicit analogy with the generation of animals divided into male and female *individuals*. That is, presuming that there must be active (animating) and passive (receiving) principles in generation, these principles are called 'male' and 'female', borrowing the terms of a distinction between kinds of *individuals* in the animal kingdom. This provides the metaphysical definition of 'male' and 'female', terms which, in their generality, are applied to all forms of life, but only insofar as their various, different senses are respected when speaking of specific instances. With a complicated self-justifying circularity, the *result* of an implicit analogical (or perhaps metaphorical) extension of the animal division between male and female individuals to the presumed active/passive metaphysical principles of generation gives us the idea of male and female principles, which are then evoked as the metaphysical basis for the analogy between animal and plant sex.

What is the justification for the characterization of the different principles or causes (of motion and matter, respectively) as 'male' and 'female'? What is the justification for the characterization of activity and passivity as male and female respectively? There is no explicit justification of this; these characterizations are, rather, the foundational presumptions of the metaphysics of 'sex'.

5 WHAT ARE ‘MALE’ AND ‘FEMALE’ IN PLANTS?

We can summarize the arguments of the previous three chapters, very briefly, as follows. Traditional histories of botany, written from the standpoint of the present, are surprised that it took philosophers, naturalists and scientists so long to discover the ‘obvious’ truth of plant sex. But from our plant philosophy perspective the hesitations and doubts of Aristotle and the thinkers of the Aristotelian botanical tradition are not historical confusions or errors; rather, they are the consequence of a properly philosophical approach to the issue. The idea of plant sex was not obvious for these thinkers because an answer to the question ‘Do plants have “male” and “female”?’ required an answer to another, more fundamental philosophical question: ‘What are “male” and “female”?’ A first response to this more fundamental question can be phrased in a characteristically Aristotelian form: ‘male’ and ‘female’ are said in many different ways. Aristotle’s distinctions between the metaphysical principles of male and female, male and female individuals and male and female parts or organs allowed these thinkers to discuss the ways in which plants could be said to have male and female: it was not a question with a ‘yes’ or ‘no’ answer. Part of this involved an explicit and critical reflection on the analogy with animals, not taking it for granted.

Whereas, then, the traditional histories of botany see the philosophical aspects of that history as a regrettable drag on scientific progress, from our plant philosophy perspective this is part of its strength. As we showed in Chapter 3, sixteenth- and seventeenth-century botany (with Cesalpino and Grew in particular) progressed partly *through*, not despite, its philosophical commitments. Concomitantly, philosophy proceeds through botany, in discussions of plant sex, because of the necessity, there, to reflect on the different meanings of male and female. The traditional histories of botany also celebrate the entry into the properly scientific era of plant science with the acceptance of the literal truth of plant sex

in the late seventeenth and eighteenth centuries, a progression which they see as dependent upon the jettisoning of the metaphysical baggage of the earlier tradition. But, as we have seen in Chapter 4, Vaillant's and Linnaeus's discussions of plant sex are in fact notable for the literal/metaphorical ambiguity of their terms, both in their descriptions and in their scientific terminology, an ambiguity which conditions and is conditioned by a collapse of the Aristotelian distinctions between the different meanings of sex, or a conflation of the different meanings. This collapse is less a jettisoning of metaphysical baggage than a philosophical regression. Because of the allegedly literal meaning now ascribed to male and female in plants, the limits and effects of the founding zoological analogy are no longer an explicit concern and the question of the being of 'male' and 'female' recedes. So, in one sense it is true that philosophy was (on the surface, at least) banished from botanical science. But what did botany thereby lose? What did philosophy lose when it apparently accepted that plants were no longer part of its intellectual domain? What if both botany and philosophy still need to ask, together: 'What do we mean by "male" and "female" in plants?'

As we mentioned in Chapter 1, the dibiontic life cycle (or 'alternation of generations'), unique to plants, complicates the identification of male and female in these forms of life. In this chapter we will look in a little more detail at the specificity of the dibiontic life cycle, which requires a brief scientific presentation. We will see how the recognition of the alternation of generations in plants once again explicitly raised the problem of the identification of male and female in plants, a problem which played out at the end of the nineteenth century and the beginning of the twentieth in a terminological debate in botany over the use of 'male' and 'female'. With this debate in mind, we will then look at some of the different usages of sexed terminology in more recent scientific and popular scientific literature on plant form and plant sexual reproduction. We will see that in this literature deviation from the strict scientific usage is surprisingly common; 'male' and 'female' are, in fact, used in several different senses, sometimes simultaneously. In some areas of plant science, indeed, a not-strictly-scientific use of the terms 'male' and 'female' is axiomatic, both textually and conceptually, as we will see. In certain respects, then, discussions of plants maintain a doubled discourse on male and female, overlaying a common-sense zoological model onto the strictly biological plant model. What are the implications of this?

In this chapter we will argue that attention to the dibiontic life cycle of plants, far from revealing anything obvious about plant sex, rather allows the full complexity of the problem to emerge. Attempts to resolve the terminological dispute about ‘male’ and ‘female’ in plants, and to propose new terminologies or definitions to describe the different sexual forms and sexual systems of plants, are, of course, based on the need to acknowledge biological reality. But they are equally – if not more so – either pragmatic compromises or attempts to construct new concepts of sex (and gender) and male and female for plants. It is this construction that is of particular interest to plant philosophy. To demonstrate this, we will focus particularly on the influential work of the botanist David G. Lloyd, from the 1970s and early 1980s. The kinds of conceptual and terminological problems and questions that Lloyd’s work addresses already arise, as we will show, in Charles Darwin’s *The Different Forms of Flowers on Plants of the Same Species* (1877), a text to which Lloyd often refers. Lloyd’s proposed botanical definitions of ‘sex’, ‘gender’ and ‘male’ and ‘female’ also arise out of an acknowledgement (although it is perhaps not wholly explicit) of the inadequacy of the animal model for the description and understanding of plant populations and their sexual reproduction. In this chapter we will argue that this kind of definitional work has an ineliminably philosophical aspect, and that Lloyd’s work on this topic, as well as being a contribution to plant science, can also be interpreted as contributing to a philosophy of plant sex.

Dibiontic life, or alternation of generations

First, though, it is necessary to appreciate the specificity of the dibiontic life cycle of plants, which is also to appreciate the specificity of the forms of sexual reproduction in plants. The phenomenon of the ‘alternation of generations’¹ in plant life cycles was first identified in 1851 by William Hofmeister in his studies of the ‘higher cryptogamia’ (mosses and ferns), drawing on his contemporaneous work on embryology in flowering plants. In 1872 Sachs described the results of Hofmeister’s investigations in plant embryology and the higher cryptogamia as ‘magnificent beyond all that has been achieved before or since in the domain of descriptive botany’. Hofmeister showed, in both flowering plants and cryptogams,

that the egg cell is formed in the embryo sac prior to fertilization and that its development into an embryo is dependent on its relation to the pollen tube. This meant, according to Sachs, that ‘the idea of what is meant by the development of a plant was suddenly and completely changed’ with alternation of generation ‘proved to be the highest law of development’, demonstrating a developmental commonality in otherwise very different plants such as liverworts, mosses, ferns and angiosperms (flowering plants).² What Sachs and others after him admired was not just the intellectual achievement of the discovery of the alternation of generations but also to some extent the unexpected and marvellous phenomenon itself. They admired, as Nils Svedelius put it, ‘that singular relation of alternation in the ontogenetic development of plants which [Carl Nägeli] called “one of the most remarkable phenomena of the plant world”’.³

The alternation of two distinct, multicellular generations in plants is now understood primarily in terms of the different ploidy of those generations, that is, the different number of sets of chromosomes. (Even the simplest presentation of this cannot avoid the fact that it is rather complex, so we must impose a couple of pages of technical vocabulary on the reader here.⁴) The haploid generation, whose cells have one copy of each chromosome, is often abbreviated as $1N$; the diploid (with cells with two sets of chromosomes) as $2N$.⁵ Before looking at this, it may be useful to be reminded of the typical animal life cycle, in contrast with which the specificity of alternation of generations in plants becomes clearer.

Animals – humans, for example – are multicellular diploid organisms; that is, their cells contain two sets of chromosomes, one set from each parent. In growth and maintenance (cell replacement) these cells divide mitotically, mitosis being a process of cell duplication (one cell divides into two genetically identical ‘daughter’ cells). The only haploid cells in the animal organism are the gametes, the ova or the spermatozoa, sometimes called the sex cells or reproductive cells. Gametes are produced (in the diploid organism) by the extraordinarily complex process of meiosis, the nuclear division (halving the number of chromosomes) that occurs after the ‘pairing’ of the maternal and paternal chromosomes in the cell and the ‘crossing over’ (swapping of DNA sequences) at specific points, or genetic recombination. The fusion of two haploid gametes (syngamy) in sexual reproduction produces a zygote, in which the diploid state is restored. The diploid zygote becomes the diploid embryo which (all being well) is born, grows, lives and dies as a diploid organism, itself usually able

to produce haploid gametes. Animal gametes, some of which are motile and ejected from the body, cannot live independently of their diploid producer and do not grow into a haploid organism. The animal life cycle is therefore said to be ‘monobiontic’ – there is only one multicellular diploid generation.

In the dibiontic life cycle of the plant, on the other hand, there are two multicellular generations, which may be more or less physically independent of each other. One generation is diploid, the other haploid.⁶ Whereas in the animal life cycle the diploid organism produces the haploid gamete by meiosis directly, in the dibiontic life cycle the diploid generation produces a haploid generation that produces the haploid gamete. The diploid generation is most often (mosses, hornworts and liverworts are exceptions) the plants that we would recognize. They are, as L. H. Bailey put it, what we would think of as ‘the plant’⁷ itself – the tree on the street, the grass in the meadow, the fern uncurling in my garden, the flowering bush, the dandelion pushing up between the cracks in the pavement. This diploid generation (or phase) is the ‘sporophyte’ (literally the ‘spore-plant’), which produces, by meiosis, spores. That ferns, for example, produce ‘spores’ is well known, not least because the spore cases (sori) are visible on the underside of leaves. It is far less obvious, because not easily seen, that the sporophyte generation of flowering plants also produce, by meiosis, microspores (in the anthers) and megaspores (in the ovule). These haploid spores divide mitotically to produce a second (haploid) generation – the ‘gametophyte’ (literally the gamete-plant) which produces (haploid) gametes by mitosis.

What is the difference between a spore and a gamete? Gametes can fuse with other gametes; indeed, with very few exceptions, if they do not undergo syngamy they die.⁸ Spores cannot fuse with other spores but they can divide mitotically and grow into a new organism – the haploid plant or gametophyte.⁹ In most cases the sporophyte and the gametophyte do not resemble each other at all; the generations are heteromorphic. Before the discovery of chromosomes, the discussion of the alternation of generations was thus at first the description of the alternate *morphologies* of the two generations, which is easy to see in ferns and mosses with the naked eye. (Ironically it is the flowering plants which are the ‘cryptogams’ in this respect.) Figure 5.1 shows, in a schematic form, the life cycle of the fern (or an idealized version of it). The sporophyte generation is the ‘the fern’ that most people would recognize; the gametophyte is represented by the smaller, heart-shaped plant. Although the gametophyte is often

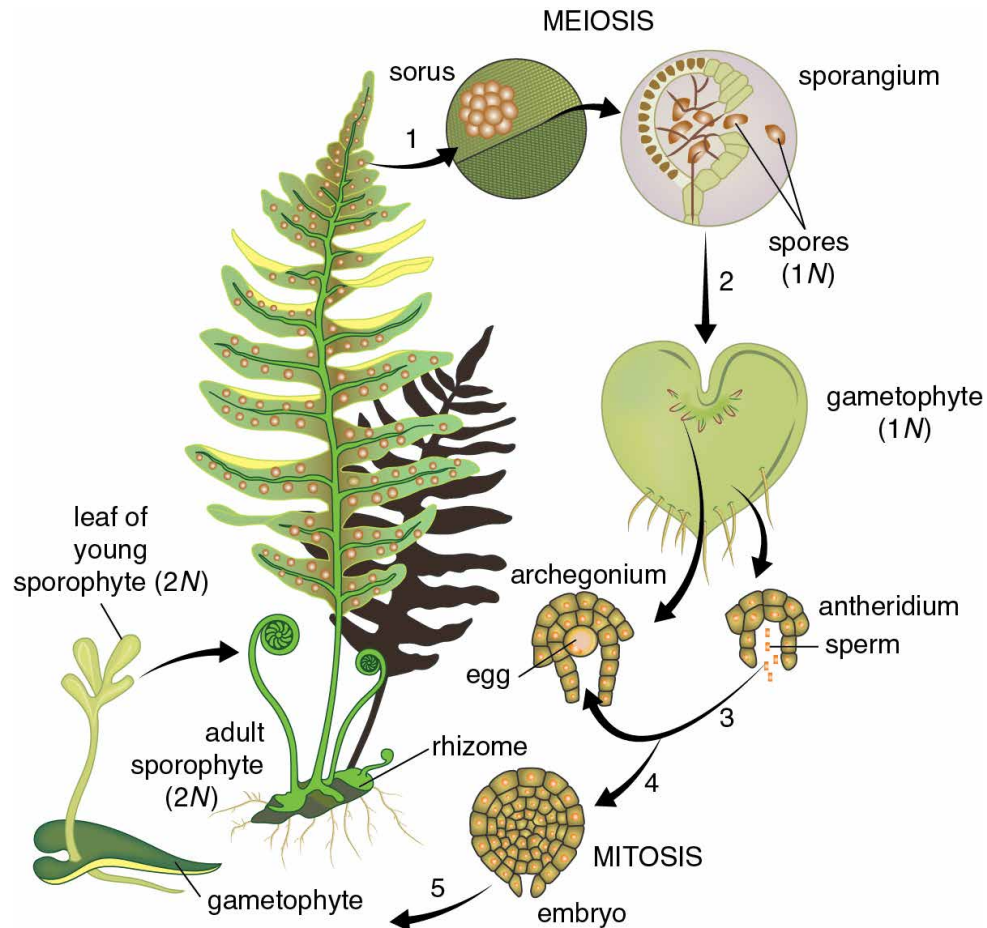


FIGURE 5.1 Life cycle of the fern. Reproduced by Integra Software Services.

very tiny, a mature gametophyte is an ‘adult’ plant. The male gametes are produced in a region of the gametophyte called the antheridium, the female in the archegonium. Ejected sperm (as they are called) swim through surface water and enter the archegonium where the eggs are produced. If fertilization occurs (gamete fusion produces a zygote that becomes an embryo) a new diploid sporophyte will grow and the haploid gametophyte dies. With mosses, on the other hand, the ‘mossy’ green plant with which we are most familiar is the haploid gametophyte phase, spreading vegetatively. The sporophyte generation is often visible as the thin hair-like stalks that grow out of and remain attached to the mossy gametophyte until they die off.

The same alternation of generations characterizes the life cycle of flowering, seed-producing plants (angiosperms) (Figure 5.2). The diploid sporophyte is the recognizable flowering plant itself. Its haploid

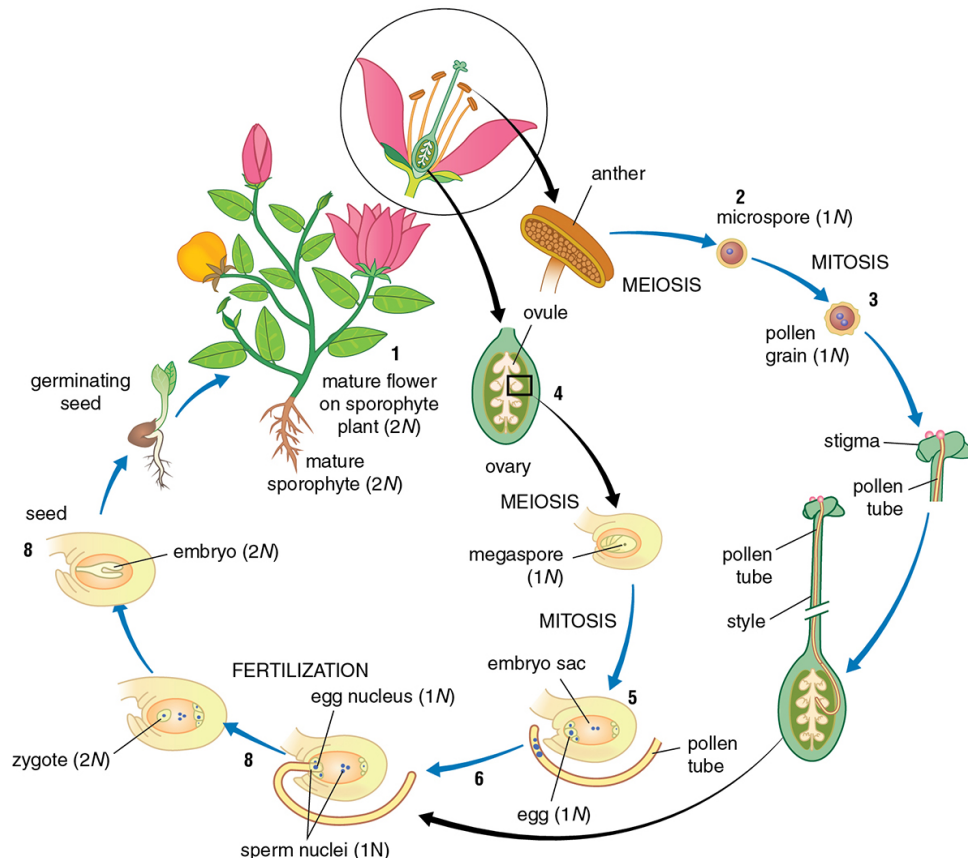


FIGURE 5.2 Angiosperm life cycle. Reproduced by Integra Software Services.

spores, the product of meiosis, are the pollen grains produced in the anthers (the microspores) and the megaspore produced in the ovule (the ovule being a part of the plant, which is itself part of the ovary; we will discuss this terminology in Chapter 6). The haploid microspores (pollen grains) develop into haploid microgametophytes comprising only two or three cells – so small that they fit inside the pollen grains. The haploid megaspore develops within the ovule into a seven-celled haploid megagametophyte for which the technical term is the ‘embryo sac’.¹⁰ The gametophytes then produce gametes by mitosis. Although it may seem counterintuitive, the microgametophyte and the megagametophyte are, like the independent gametophytes of ferns, entire and mature plants – a distinct, multicellular generation from the sporophytes which produced them. The haploid angiosperm gametophyte is ‘a tiny mass of cells with no roots, stems, leaves or vascular tissues, but it is an entire plant’. And this is the case even though the megagametophyte develops *within* the sporophyte plant – it is ‘one plant growing inside another’.¹¹

The microgametophyte contains two sperm cells or microgametes. If a viable pollen grain from an anther arrives on a stigma it may germinate to produce this gametophyte generation within itself (within the pollen grain) and produce a pollen tube that grows rapidly down through the style, carrying the sperm cells to the ovule and then to the megaspore, which develops into the megagametophyte. One sperm cell (microgamete) and the egg cell (or ‘ovum’; the megagamete) undergo syngamy to produce the diploid zygote that becomes the embryo and will eventually grow into the next generation (diploid) sporophyte. The second sperm cell fuses with the binucleate ‘central cell’ of the megagametophyte to produce the triploid endosperm. Because two sperm cells undergo fusion with different cells from the megagametophyte, the process is known as ‘double fertilization’.¹² The life cycle of ‘the’ flowering plant and ‘the’ fern or moss thus involves at least three or four distinct multicellular organisms – minimally one sporophyte and two gametophytes¹³ – but the sporophyte (a tree, for example) may, via the vast number of spores from its many hundreds of flowers, give rise to many hundreds of gametophytes in each flowering season.

These are the basics of plant sexual reproduction, the details of which can be found in any textbook of botany and in multiple iterations online.¹⁴ What interest might plant philosophy have in this?

‘On the untechnical terminology of the sex-relation in plants’

The fact of the alternation of generations in plants raises, once again, the question of what we mean by ‘male’ and ‘female’ in plants, and once again refers us to the prior question: ‘What are “male” and “female” more generally?’ Almost from the moment of its first description, the alternation of generations in plants was characterized as an alternation of asexual and sexual generations. The sporophyte generation was (and sometimes still is) called the asexual generation because its spores (unlike seeds) are produced asexually (i.e. without conjugation of different cells) and because they give rise to a new plant without themselves being fertilized. The spore itself was thus also called asexual.¹⁵ The gametophyte (as it is now known) is sometimes called the sexual generation because it produces the gametes through which plants reproduce sexually.¹⁶ This

means that the sexual generation arises from an asexual generation. It also means that, strictly speaking, neither the sporophyte (what we mostly recognize as ‘the plant’ or ‘the tree’) nor any of its structures – including the structures of the flower(s) – are ‘male’ or ‘female’.

In 1896, after the discovery of chromosomes and the introduction of the terminology of ‘gametes’, the influential American horticulturalist L. H. Bailey reflected on the implications of the sporophyte generation being understood as asexual. He noted that the discovery of the alternation of generations and the new conception of the ‘sex-relation’ in plants had given rise to new terminology and to attempts to ‘restrict or to specialize the use of such age-long words as male and female, sex and the like, when applying them to plants’. Bailey reminds the reader that the ‘original conceptions of sexuality in plants... were borrowed and adapted very largely from analogy with the animal kingdom’. This analogy, he points out, gave rise to more than just the idea of male and female in general: ‘The stamens were considered to be male organs of sex and the pistils to be female organs, the idea of the necessity of a conformed sex-member being evidently borrowed from a knowledge of animal morphology.’¹⁷ Hofmeister’s discovery of a ‘sex-bearing’ generation followed by a ‘sexless’ generation has thus raised the questions, Bailey says, of ‘how far can we use the terms “male” and “female” and what must be the common language of the sex-relation in plants?’ As the ‘male sex phase’ is essentially only the ‘short span and function of the generative cell developing from the pollen grain’ and the female the development of the gametophyte in the ovule, some now object to calling the stamen ‘male’ and the pistil ‘female’. Bailey sees that this objection is scientifically and logically warranted. Indeed, he says that if we carry this scientific logic to its end, we would not call any part of the gametophyte except the egg cell and the sperm cell ‘female’ or ‘male’: ‘it follows that we cannot apply the terms “male”, “female”, “sex”, and the like, to plants, save in the very brief period during which impregnation is taking place.’¹⁸ And what, he says, if this ‘hypercriticism’ were applied to animals too? There would be ‘pandemonium’, as ‘One could not speak of the members of generation as sex organs, not of any animal as male or female.’¹⁹

Bailey’s overwrought fears of pandemonium are interesting, and no doubt betray something of the broader ideological context in 1896. But when botanists (and especially those concerned with botanical education) did rather straightforwardly seek to reflect the existence of

the asexual generation in their terminology the sky did not fall in. The American botanist and educator John Merle Coulter, in various works, made a concerted effort to correct (as he saw it) the misnaming of the parts of flowers as 'male' and 'female' and it is notable that he did this in textbooks and shorter books intended for students, including secondary (or high school) students. In his *The Evolution of Sex in Plants* (1914) he defines alternation of generations as the separation of a spore-producing and a gamete-producing generation:

so that there are two kinds of individual in a life cycle.... The individual with the sex organs (gametophyte) may be called the sexual individual, and the one without sex organs (sporophyte) may be called the sexless individual. Alternation of generations, therefore, is the alternation of a sexual individual with a sexless individual in a life cycle. This means that it takes two individuals to complete a life cycle.²⁰

The evolution of gamete production is at the same time, for Coulter, the evolution of sex, in the sense of the 'act of fusion' and the evolution of what will be identified as 'the sexes'.²¹

His discussion of this includes an implicit reflection on the meaning of 'male' and 'female' in general. The first (evolutionary) stage of gamete production, he notes, is called 'isogamy', a term which is justified to the extent that it refers to the morphological likeness of the two gametes, but there must also be a physiological differentiation. Isogamy 'shades into' heterogamy which 'must not be regarded as the gradual differentiation of *maleness* and *femaleness*, but the gradual appearance of differences that make the two sexes recognisable'.²² 'At this stage in our knowledge', he writes, 'it is useless to ask what "maleness" and "femaleness" are, but it seems evident that they represent essentially conditions of the nucleus'. That is, we *call* heterogametes 'male' and 'female', but these are 'words rather than explanations', standing in for 'a definite physiological difference'.²³

Having located the distinction of sex at the level of the (sex) cell, Coulter also describes the evolution of the sex organs of plants (the 'gametangia') as having moved through three stages: unprotected gamete-producing cells associated with water habitat (in algae) jacketed organs (antheridia and archegonia) associated with the land habitat (e.g. ferns) and the 'final elimination of antheridia and archegonia, associated with the complete dependence and reduction of the male and female

plants [i.e. of the gametophytes]. In flowering plants, the archegonia (that part of the fern gametophyte that produces the female gametes) is 'reduced to its essential sexual structure, the egg'.²⁴ It is very specifically the sexual *organs* of both gymnosperms and angiosperms that Coulter describes as having been 'completely reduced' (or indeed, in the case of the female organ, 'eliminated').²⁵ If we can speak of the sexual organs – that is, the gamete-producing cells – of flowering plants, for example, we must be referring to a part of the gametophyte or to the gametophyte itself. As the gametophytes of flowering plants have 'disappeared from ordinary observation' in becoming very tiny and encased in the spore or the sporophyte, 'the sex organs can only be discovered by the use of laboratory technique'.²⁶ Coulter writes that it is therefore no surprise that earlier botanists, who worked out that the stamens and pistils were both necessary in the life cycle of the plant, *misidentified them* as the sexual organs of the plants. However, 'to regard a flower as a sex structure and its stamens and pistils as sex organs is to misapprehend the situation. They belong to the sporophyte, which does not produce sex organs... To speak of male flowers and female flowers, as is so often done, is natural, but it is untrue'.²⁷

Coulter insists on calling the structures that produce spores 'sporophylls' (a term that today is sometimes defined as referring only to the spore-bearing leaves of ferns and mosses): 'Before the real nature of these structures was known, the microsporophyll was called a *stamen*; the region of the microsporophyll bearing sporangia was called an *anther*; the microsporangia, *pollen sacs*; and the microspores, *pollen grains*'.²⁸ In his *Plant Studies* (1901), a textbook for secondary school students, he had already noted that 'stamen', 'pollen-sacs', 'pollen-grain' and 'pollen' were the names given to the structures of seed-plants before their function was understood, although there he also says that they are 'names... still very convenient to use in connection with the Spermatophytes' so long as we remember that they are really microsporophylls.²⁹ However, there is no justification for referring to cones or to stamens and pistils as 'male' or 'female' organs, as far as Coulter is concerned, not even for convenience – it is simply biologically wrong.³⁰ As an educationalist, Coulter objected to the false teaching involved in referring to the 'sexual organs' of flowers: 'It should be borne distinctly in mind that the stamen is not a sex organ, for the literature of botany is full of this old assumption, and the beginner is in danger of becoming confused and of forgetting that pollen grains are asexual spores'.³¹

Bailey, by contrast, claims that the fact that the pollen grain and its gametophyte ultimately come from the stamen means that the botanist continues to be warranted in calling the stamen 'male', and recommends the retention of the 'old-time attributes' of maleness and femaleness for the parts of the flower:

'Male' and 'female' never did and never can be made to express strict morphological homologies. An organ of an animal or a plant is male if it exercises the functions of paternity and not of maternity. The stamen is such an organ. Its entire office is that of maleness... The common language of sex has always dealt in analogies. There are perfectly good and sufficient technical terms to designate the homologies and the ultimate physiological processes.³²

While acknowledging the strict scientific correctness of the restriction of the uses of the terms 'male' and 'female' to gametes, Bailey thus argues that we should nevertheless retain the 'untechnical' terminology, borrowed from zoology and analogically understood, for the parts of flowers, individual flowers and individual plants. In 1905 Charles J. Chamberlain made much the same argument, again explicitly acknowledging the technical correctness of restricting 'male' and 'female' to gametophytes but recommending that the botanist should follow the zoologists' practice and to continue to speak analogically of male and female sexual organs in the flower and male and female individuals.³³

And so it has come to pass in botany, to this day. In the long century since Bailey's article the botanical consensus is that the sporophyte and the parts of the flower are not, strictly speaking, 'male' or 'female' because the more general biological definition of male and female has come to be restricted to the designation of anisogamy (having two gametes of unequal size) and to the organisms that produce such anisogametes.³⁴ Only the gametophyte qualifies. Nevertheless, in almost all scientific work on plant reproduction, and in its popular communication, the zoological analogy continues to hold sway and sporophytes and the parts of flowers are routinely designated 'male' and 'female' (and, of course, hermaphroditic). We should have to say, then, that the much-vaunted progress of botany from an analogical to a literal understanding of male and female in plants, as discussed in Chapter 4, is not what it seems, and the dominant way of speaking about plant sex is only misleadingly characterized as 'literally true'.

There are examples of strict attempts to stay faithful to technical biological definitions of male and female in discussions of plants. James Mauseth's widely used botany textbook explains that the parts of the flowers are not, strictly speaking, male and female and sticks thereafter to a technical – that is, specifically botanical – vocabulary that avoids the explicit zoological analogy. Monoecy and dioecy are explained without reference to male and female, flowers are called 'staminate', 'carpellate', 'imperfect' and 'perfect', and (just as Coulter wished) stamens and carpels are referred to as 'microsporophylls' and 'megasporophylls', respectively.³⁵ Mauseth uses the anthropomorphic language bequeathed by Linnaeus only where it has become established in technical terms of botanical art required by the student of botany, referring, for example, to the 'androecium' (the collective term for the stamens) and the 'gynoecium' (the carpel, including stigma, style and ovary).³⁶

It is also possible to find examples of discussions of specific flower structures, usually where the concentration is on only one structure, that avoid all mention of male and female.³⁷ But far more commonly – to the point that we can say that the practice is hegemonic – plant scientists (and, in their trail, the authors of popular scientific literature and of books for children) refer freely to the stamens/anthers and pistils/ovaries as the male and female sexual organs of the plant, and to male and female flowers and male and female (sporophytic) plants. Reference to the stamens, for example, as the male sexual organ is characteristic of the scientific literature that deals with the reproduction of flowering plants at the highest level of generality, with the least physiological detail, and with no mention of the alternation of generations.³⁸ It is also characteristic of the popular scientific literature and the children's literature on plants. It is perhaps not surprising to find some bowdlerization in the latter, especially that aimed at primary school-aged children. But the pedagogic constraints of children's literature do not extend to the popular scientific literature aimed at adults, which nevertheless often reproduces about plant sex only what one may find in books intended for, say, nine-year-olds.³⁹ This is the case even in the plant advocacy literature discussed in Chapter 1, which otherwise aims to introduce the reader to cutting-edge research in plant science and presumes on the part of the reader a willingness to follow more detailed scientific explanations in other areas.⁴⁰

In the scientific literature it is not uncommon for authors to acknowledge, explicitly, that this usage is, strictly speaking, incorrect, before they proceed to employ it.⁴¹ Others acknowledge it implicitly,

as when Attenborough writes that ‘pollen is conventionally described as being a flower’s male cells, and the stamens, which produce it, as the male organs’.⁴² But what is perhaps most striking is the widespread and often unremarked use of this ‘strictly speaking incorrect’ terminology in the field where one might least expect to find it – precisely that of plant sexual reproduction. Where that literature deals with plant cell biology (physiology) and genetics below the level of the flower part, the whole flower or the whole plant, discussion can proceed without any mention of male or female. But where the concern is with populations and mating systems (or sexual systems), plant breeding and evolution, which involves consideration of the gross morphological and quantitative aspects of ovule and pollen production and of floral traits, of individuals and groups of plants, talk of male and female organs, flowers and individuals is commonplace. In other words, it is perhaps where plant sexual ‘behaviour’ is at issue that the ‘old-time’ terminology of male and female reasserts itself most insistently.⁴³

‘The whole case appears to me a very curious one’: Darwin on plant sex

However, it was also in this field (plant sexual reproduction) that the most theoretically sophisticated reflection on the concepts used to describe plant sex arose, in the work of the influential New Zealand botanist David G. Lloyd. In a number of papers (some co-authored) on plant sex ratios, breeding systems and sexual strategies in specific plants, Lloyd progressively refines the terminology used to refer to the different ‘sexes’ of flowers. Lloyd’s concerns are botanical and not explicitly philosophical but, throughout, the texts implicitly and explicitly ask one of the central questions of the plant philosophy tradition: ‘What do we mean by “male” and “female” in plants?’ Lloyd’s work may thus be read *as* or at least *through* plant philosophy and doing so yields a result that further complicates both the traditional history of the theory of plant sexuality and the idea that botany becomes properly scientific in understanding plant sex in a ‘literal’ sense. Lloyd can be read as *constructing* the concepts of ‘male’, ‘female’, ‘sex’ and ‘gender’ for specifically botanical purposes: that is, concepts more adequate to the specificity of plant life. The meaning of these concepts, for Lloyd, is not read off what is given to the botanical gaze; it is not simply empirically derived. Further, Lloyd’s concepts of ‘male’, ‘female’, ‘sex’ and ‘gender’ are at odds with some, if

not all, of the ways that the same words are used in relation to animals. This is because Lloyd's achievement is to have worked these concepts for plants specifically in such a way that they can accommodate both the strictly biological definitions (advocated by Coulter, for example) and the 'old-time' terminology, but without, in the end, staking a claim to have arrived at the final truth of plant sex. If it is nevertheless possible to retain some general biological concepts of 'male' and 'female' and sex, across all relevant kingdoms of life, we will thus find them functioning within a complex field of different concepts of 'male' and 'female' and 'sex' (and 'gender'), giving the lie to the idea that there is anything obvious or simple about sex for any form of life. (We attempt to give a positive philosophical account of that claim in the last part of this book, in the Epilogue.)

An obituary of Lloyd describes him in his early career (1965–75) as 'making observations on natural history in the Darwinian tradition'.⁴⁴ However, Charles Darwin's acknowledgement of the complexity of the 'sexual' forms of flowers and his admitted perplexity about these forms can also be understood as the basis of Lloyd's subsequent theoretical reflections on plant sex. In *The Different Form of Flowers on Plants of the Same Species* (1877) Darwin considers certain forms of 'sexual relations' in plants – what are now called 'breeding systems', 'mating systems' or 'sexual systems',⁴⁵ that is, the characterization of the species or sub-specific group in terms of the distribution of sexuality across flowers and individuals. He uses some of Linnaeus's basic terminology, distinguishing between monoecious, dioecious, hermaphrodite and polygamous species – that is, those with male and female flowers on the same individual (monoecious), those with male and female flowers on different individuals (dioecious), those with male and female in the same flower (hermaphrodite) and those with male, female and hermaphrodite flowers on the same or different individuals (polygamous). With the meaning of the term 'polygamy' thus restricted, he introduces four other categories: the gynomonoecious (with female and hermaphrodite flowers on the same individual), andromonoecious (with male and hermaphrodite flowers on the same individual); and the gynodioecious (with female and/or hermaphrodite flowers on different individuals) and androdioecious (with male and/or hermaphrodite flowers on the different individuals).⁴⁶ Already we can see that the application of the terms 'male' and 'female' and the presumption of the universality of the male/female binary at the level of the flower and the individual (which is already muddled by the existence of the hermaphrodite: Is the hermaphrodite itself a third sex?) cannot be

straightforwardly maintained in the discussion of these new categories. In the gynomonoecious species, for example, each individual is both female and hermaphrodite; in the gynodioecious species each individual is either female or female/hermaphrodite. In these species there seems to be no 'male' at the level of the individual. However, Darwin also introduces an important distinction between form and function, noting that there may be flowers that are ostensibly morphologically hermaphrodite while being impaired in one or other function – for example, morphologically hermaphrodite flowers which in fact produce no seed, thereby being 'in function a male'.⁴⁷ This distinction between morphology and function will furnish the basis of Lloyd's conceptual innovations in the thinking of plant sex and, as we shall see, it quite upsets the idea that plant sex is something obvious.

Darwin's main concern in his 1877 text is with the dimorphic and trimorphic sub-groups of the hermaphrodite class. He calls these 'heterostyled plants', that is, plants with two or three forms of hermaphrodite flowers, with a combination of different sized styles (or pistils) and different sized stamens in each. Two main types of these engage Darwin's attention. First, there is *Primula veris* (Cowslip), which has two forms, each with differently sized pistils and stamens. The two forms are known as the 'long-styled' (also called 'pin') plants and the 'short-styled' (also called 'thrum') plants (see Figure 5.3). In the discussion of this species Darwin recounts that, in asking himself about the 'meaning' of the existence of these two forms, he first thought that the species must be evolving towards dioecy and presupposed 'that the long-styled plants, with their longer pistils, rougher stigmas, and smaller pollen grains, were more feminine in nature, and would produce more seed' and that the short-styled were more 'masculine'. But, contrary to his expectations, collections of seeds from both showed the latter to be the more fertile, which for him meant that the short-styled proved to be more feminine in nature.⁴⁸ It is notable that he here introduces (and continues to maintain) a terminological distinction between 'male' and 'female' and 'masculine' and 'feminine'.

In further experiments Darwin discovered that long-styled flowers pollinated with pollen from the short-styled form were more fertile (i.e. they produced more seeds) than those self-pollinated or pollinated from other long-styled plants. Conversely, the short-styled form was most fertile with pollination from the long-styled form.⁴⁹ The grades of fertility in individuals could range from zero (completely infertile) to

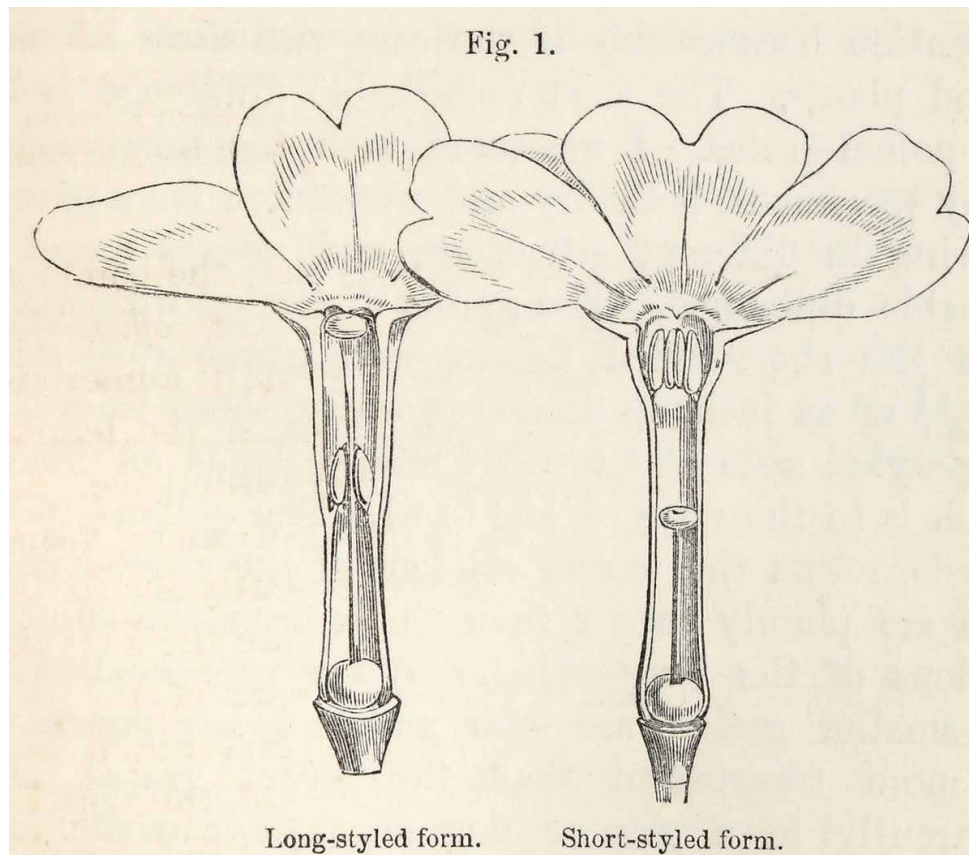


FIGURE 5.3 Illustration of dimorphic heterostyly in *Primula veris*, from Darwin's *The Different Forms of Flowers*. We can see that both the style and the stamens are of different sizes in the two forms. By permission of the Linnean Society of London.

‘perfect’ or ‘full’ fertility, with all stages between. With perfect fertility as the ideal, Darwin then reconceives the sexual nature of the dimorphic heterostyled species:

The individual plants of the present species [*Primula veris*]... are divided into two sets or bodies, which cannot be called distinct sexes, for both are hermaphrodites; yet they are to a certain extent sexually distinct, for they require reciprocal union for perfect fertility. As quadrupeds are divided into nearly equal bodies of different sexes, so here we have two bodies, approximately equal in number, differing in their sexual powers and related to each other like males and females.⁵⁰

As the use of the animal analogy betrays, Darwin is searching here for a binary that is not in fact supported by the facts as he finds them.

For, according to his own lights, those who are 'like males' can fertilize others who are 'like males', and those who are 'like females' can fertilize others who are 'like females', even if they are less fertile in so doing. But the important point is that the male and female *function* is once again being conceived – albeit confusedly – as distinct from male and female morphology. This function is also expressed in terms of 'sexual powers'. Later, in relation to another dimorphic heterostyled species, *Linum grandiflorum* (known as Flowering Flax, amongst other names), Darwin's experiments seemed to show that the long-styled form was absolutely sterile when pollinated with own-form pollen and the short-styled less fertile when pollinated with own-form pollen.⁵¹ As the pollen grains from both forms were undistinguishable under Darwin's microscope, the nature of the 'power' of the stigmas to cause the pollen tube to grow from other-form pollen was to him deeply mysterious. But it is established that the two pollens and the two stigmas clearly 'mutually recognise each other by some means',⁵² and that it is what he later calls the 'sexual elements' or 'reproductive powers' that determines this, not 'any general difference in constitution or structure'.⁵³ Darwin thus distinguishes the male and female 'fertilizing power' from the form of the male and female flower or male and female individual, evoking the Aristotelian distinctions between male and female principles, male and female organs and male and female individuals – or at least highlighting, once again, the need to revive the question of what is meant by 'male' and 'female' in each case.

The second type of heterostyled plants that Darwin discusses complicates things even further. *Lythrum salicaria* (Purple Loosestrife) is trimorphic (see Figure 5.4). The manner of fertilization of this plant, according to Darwin, offers 'a more remarkable case than can be found in any other plant or animal'. In his own description:

The pistil of each form differs from that in either of the other forms, and in each there are two sets of stamens different in appearance and function. But one set of stamens in each form corresponds with a set in one of the other two forms. Altogether this one species includes three females or female organs [long-styled, mid-styled and short-styled] and three sets of male organs [combinations of 'longest', 'mid-length' and 'shortest'], all as distinct from one another as if they belonged to different species; and if smaller functional differences are considered, there are five distinct sets of males.⁵⁴

Fig. 10.

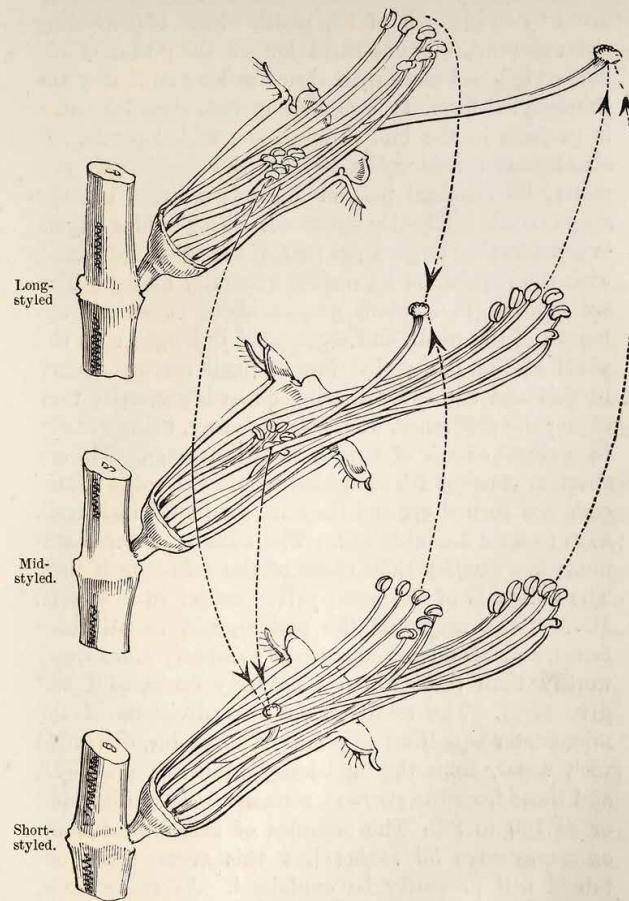


Diagram of the flowers of the three forms of *Lythrum salicaria*, in their natural position, with the petals and calyx removed on the near side: enlarged six times. The dotted lines with the arrows show the directions in which pollen must be carried to each stigma to ensure full fertility.

FIGURE 5.4 The three forms of *Lythrum salicaria*, from Darwin's *The Different Forms of Flowers*. By permission of the Linnean Society of London.

Darwin's experiments showed him that the greatest fertility is achieved when pollen from the longest stamens pollinate the longest pistil, when pollen from the mid-length stamens pollinate the mid-length pistil and when pollen from the shortest stamens pollinate the shortest pistil – these are what he calls the 'legitimate unions'.⁵⁵ However, most of the 'illegitimate' unions also resulted in seed, albeit in much reduced number. The fact that these experiments required Darwin to make 'eighteen distinct unions in order to ascertain the relative fertilizing power of the three forms' shows 'the extraordinary complexity of the reproductive system of this plant' for which, he says, 'nature has ordained

a most complex marriage-arrangement, namely, a triple union between three hermaphrodites'.⁵⁶ He also found the mid-styled form to have the highest capacity to be fertilized but the lowest pollen potency. He cites previous experiments which showed that this form also has the smallest pollen grains. From this he concludes that the mid-styled form 'appears to be highly feminine in nature' and, as with *Primula veris*, his speculative instinct is to suppose that this form is tending towards becoming fully female. However, he also notes that the female organs of this form are already fully fertile, meaning that they are in effect already fully female. At the same time, because its stamens produce an abundance of pollen, it is not possible to consider them as being in a rudimentary condition. It is both fully female, then, and at least partially male. 'The whole case', he says, 'appears to me a very curious one'.⁵⁷

It is clear that what is 'curious', for Darwin, is not the mere fact of differential fertility rates but the impossibility of a clear distinction between male and female and the combinations of different sexual organs of the same 'type', such that the same species is found 'bearing three females, different in structure and function, and three or even five sets... of males; each set likewise consisting of half-a-dozen [stamens], which likewise differ from one another in structure and function'.⁵⁸ Multiple 'sexual organs' in a flower is of course nothing unusual; almost all flowering plants have more than one stamen, for example. But *Lythrum salicaria* presents Darwin with three different *types* of females and up to five different *types* of males. The basic sex duality of male-female is here replaced with a sex multiplicity of female1-female2-female3-male1-male2-male3-male4-male5. Moreover, some females are more feminine than others, and some males more masculine, suggesting not a sex duality but a sex continuum, even at the level of the organ. Any straightforward transfer of the sex duality male-female from organ to flower to individual is thus thoroughly disrupted. The differential fertility of dimorphic heterostyled plants, some of which are infertile with own-form pollen, allows Darwin to *project* a sex duality on them,⁵⁹ but *Lythrum salicaria* resists even this and forces the question of what we mean by 'male' and 'female' at different levels (sexual organ, flower, individual) and of the relationship between these levels. Basically, the conflation of the different meanings of male and female that is both the presupposition and the result of Linnaeus's sexual system – much of the terminology of which Darwin assumes – here begs to be disaggregated and rethought. The application of the basic zoological model of sex to plants – in which one

does *not* ask ‘What are “male” and “female”?’ – is revealed to be not only inadequate but also confusing (both conceptually and terminologically) in trying to understand the specificity of vegetal sex. Plants appear as sexual oddities as compared with animals rather than as manifesting ‘normal’ forms of a specifically vegetal sexuality. The plants need, instead, a plant model. This is what David Lloyd attempted to provide.

The plant model of gender: David G. Lloyd and the sex/gender distinction in plants

In the earliest of the essays with which we are concerned here, one feature of Lloyd’s work across the next decade already emerges: a frustration with what he will later call ‘typological’ approaches to the description of sexual systems (also called ‘breeding systems’ or ‘mating systems’) in plants. A ‘sexual system’, as we noted earlier, is the characterization of the species or sub-specific group in terms of the distribution of sexuality, or ‘sex expression’,⁶⁰ across flowers and individuals. The idea of ‘sex expression’ is already ambiguous, referring both to the sex expressed in any particular individual or flower and to the characteristic forms of such expressions in any given species. It is the latter which is a sexual system. Lloyd criticizes the traditional mode of identification and definition of such systems, that is, the traditional morphological *method* of identification and the typological *result* of that method, although the form of the result is already contained in the method (the search for morphological types or a distribution of types). The probably unwitting presupposition of this is the idea of a fixed set of possible sexes and possible sexual systems. This was already being questioned in Darwin’s *The Different Forms of Flowers on Plants of the Same Species*, where, as previously discussed, he implicitly raises the idea of a sex continuum, but the idea was still strong enough in 1972 for Lloyd to feel the need to argue against it. This is obviously not because any plant scientist in fact held an explicitly static view (a significant amount of the literature on sexual systems is concerned with, for example, the evolution of dioecy from hermaphroditism); it concerns the implicit effects of the continued use of a terminology derived from a Linnean-style typological impulse and the lack of any determined alternative principle and method for the identification of sexes and sexual systems in plants. In work from 1972 to 1980 Lloyd progressively offered such an alternative.

The identification of sexual systems is obviously bound up with the identification or definition of sexes. In introducing an alternative

conception of plant sex, Lloyd switches, quite deliberately it seems, to the term 'gender', with some essays effectively reserving 'sex' for the older morphological conception.⁶¹ However, he also frequently speaks of the older model in terms of 'morphological' or 'phenotypic' gender, which he associates, somewhat mysteriously, with 'verbal descriptions'. The identification and description of the 'phenotypic gender' of an individual plant is 'based on the reproductive characteristics of that plant alone – counts of the percentage of ovuliferous, seminiferous, or polleniferous flowers, for example'.⁶² Thus the identification of 'phenotypic gender' is centred on the individual plant, in isolation from others of its species, with the visible sex 'organs' playing the primary role: in effect, organ morphology is plant sex. This is the presupposition of the 'old-time' terminology that, while strictly speaking incorrect, is still widely used.

Lloyd does not object to the old-time terminology on the basis of its being strictly speaking incorrect; so what, for him, is the problem with this morphological conception of plant sex? According to Lloyd it is that it tends to presuppose neat divisions between sexual systems and the sexual expression of individual flowers and plants, such that exceptions to the neat divisions (e.g. the appearance of hermaphroditic flowers in dioecious populations, pollen-bearing flowers on otherwise 'female' plants, polleniferous parts in otherwise 'female' flowers and so on) are 'dismissed as unimportant anomalies, irregularities, or "intersexes": that is, as freaks'.⁶³ Just as the mammal hermaphrodite or intersexed individual is seen as an anomaly or irregularity, deviating from the strict male-female dichotomy, so the polleniferous female plant is seen as an irregularity deviating from the strict, morphologically derived male-female-hermaphrodite trichotomy. But what if the freak is normal? What if the neat typology is the problem?

Substituting a functional criterion for the morphological one, Lloyd proposes a set of definitions that can accommodate all irregularities. However, they do not do this by multiplying the number of recognized gender classes or sexual systems.⁶⁴ Lloyd's is not an MFH+ approach. Instead, Lloyd's functional criterion for plant gender maintains the traditional or old-time categories of male and female but redefines them in an inclusive manner (there are no freaks; no plant is an exception) that allows for – indeed expects – the gender expression of some individuals and of populations to change over time. It is a generous concept of gender. Although Lloyd begins to put a functional concept of plant gender to

work as early as 1972, the fullest theoretical expression of this conceptual innovation is found in two essays from 1980. These lay out two main aspects to the functional concept of plant gender.

First, plant gender is quantitative. The open system of plant growth and its modular form mean that the 'sexual organs' are typically repeated many times in any given individual. Whether the 'descriptive units of sex' are sporophylls, sporangia or gametes, 'the relative proportions of male and female organs may vary over the entire range between the all-male and all-female extremes. The intermediate conditions are often fully functional. Detailed descriptions of plant gender should therefore be quantitative'.⁶⁵ The functional, quantitative conception of plant gender measures the relative paternal and maternal investments of a plant, calculating 'the proportions of a plant's genes that are most likely to be transmitted through pollen (its maleness) or through its own seeds (its femaleness), based on expenditure on androecia and gynoecia'.⁶⁶ Calculating the gender of a plant – its functional maleness or functional femaleness – is based on pollen, ovule or seed counts, or on the number of flowers bearing pollen, ovules or seeds. It thus still relies on the observational methods of the natural history tradition and of Darwinian botany. It further accepts that all calculation of plant gender will be an estimation.⁶⁷ Sometimes, notably in the case of dioecious populations with little internal variation, the functional gender will match the morphological sex of the traditional method: plants bearing exclusively gynoecia are female; those bearing exclusively androecia are male.⁶⁸ But often, especially in the case of gynodioecious populations where the morphological approach identifies females and hermaphrodites, the functional designation of gender will be at odds with the morphological designation. In a gynodioecious population those plants with exclusively gynoecia are functionally female because the plant's genes are obviously most likely to be transmitted through the maternal function. In the same population a plant with both gynoecia and androecia is more likely to transmit its genes through the paternal function, so it is described as functionally male,⁶⁹ even if it has functional gynoecia and even if it turns out that any given individual or flower *in fact* performs a female function, that is, is pollinated and fertilized. This is perhaps the most counterintuitive conclusion from the point of view of the traditional, morphological conception of plant sex: males may have functional female organs and males may be fertilized by other males and bear fruit and seed.⁷⁰ Similarly, in an androdioecious population, functional

females may and probably will have functional androecia and fertilize other functional females. There is no contradiction, therefore, in the idea that some males perform a female function or vice versa: 'Maleness and femaleness in plants are recognized as quantitative phenomena, not absolute distinctions.'⁷¹ There is a 'gradation rather than a discontinuity in gender values.'⁷²

The second main aspect of Lloyd's functional definition of gender is that 'it has no meaning for single plants considered apart from the population in which they would interbreed.'⁷³ Whereas the morphological (phenotypic) conception of gender *only* considers single plants (or flowers), the functional conception of gender 'considers a plant's sexual performance in relation to that of the other individuals in the population.'⁷⁴ Whether any plant with both gynoecia and androecia will be male, female or 'co-sexual'⁷⁵ depends on the other plants in that population. The morphology of gynoecia and androecia alone cannot determine it except in the case of strictly unisexual populations (where individuals can only produce either male or female gametes, never both), and even then occasional 'hermaphrodite' flowers of plants will still be classed as either male or female (whereas in a monomorphic population with only such flowers, they would be co-sexual).

Together, these two aspects of Lloyd's functional concept of gender mean that the identification of any plant or flower (or any 'morph')⁷⁶ as 'male', 'female' or 'co-sexual' is not a static identification of the innate sexual being or nature of any given individual or flower but a snapshot within a dynamic population, especially where – as is common – the different sexual systems and genders shade into each other.⁷⁷ As Lloyd says in 1979, 'the terms "cosex", "male", and "female"... refer to the average performance of a morph and do not specify the limits of behaviour of the individuals of that morph.'⁷⁸ As this concept of gender is based on potential, not actual function, the identification of plant gender is also in many cases speculative. Lloyd acknowledges that the complete estimate of the functional gender of plants – which requires maternal and paternal fitness to be measured through gamete production, fertilization, seed maturation and dispersal, and the growth of offspring to adulthood – is 'a formidable or even impossible task.'⁷⁹ Nevertheless, incomplete estimates are valuable in determining the likely relative success of a plant as a maternal or paternal parent. Again, this stresses that it is the sexual strategies of populations of a given species that determine the identification of the gender of any individual plant, marking a significant break from

the individual-centred morphological approach and definitively casting aside the idea that there is anything obvious about plant sex.

The relational epistemology of vegetal sex

What is the relation of Lloyd's theoretical work on the concept of plant gender to the terminological disputes about plant sex that we discussed earlier in this chapter? The positions in those disputes seemed to be polarized between two alternatives: on the one hand, an insistence on the technically correct restriction of the idea of plant sex to gametes and gametophytes (as in Coulter); on the other, a plea for the continued, common-sensical and analogical use of the old-time, 'untechnical' terminology of male and female to refer to the parts of flowers and to individual plants, that is, extended metonymically to the sporophyte generation (as in Bailey). But Lloyd's work does not accord with either of these alternatives; rather, it supersedes them in proposing new, specifically vegetal definitions of male and female and distinguishing between the concepts of sex and plant gender.

The quantitative concept of plant *gender* privileges the population over the individual and function over morphology and in so doing redefines the concepts of male and female *for plants*. This does not deny the basis of the definitions of male and female in the gamete, but it does refuse the progression of definition, implicit in the morphological individual conception of plant sex, that sees the gamete as defining the organ and the organ as defining the sex of the flower and/or plant. Lloyd's is a relational conception of gender that thinks gametes (represented in investments in gynoeceia and androeceia) in the individual plant or flower *only* insofar as they are subordinated to their potential distribution within a population. The fact of the potential or actual production of female gametes does not necessarily render a flower or plant female or hermaphrodite; rather, the total maleness or femaleness of a population determines the gender of the individual. Similarly, the identification of a morph as male or female attributes to it an affinity with a population class; it does not specify its individual nature.

As we have said, Lloyd's use of the term 'gender' seems deliberate, marking the conceptual innovation of his work. And although he continues to use words and phrases such as 'sex expression', 'sex ratios', 'cosexual', 'sexual class' and so on, the theoretical thrust of the work is towards a distinction between functional plant *gender* and the traditional

understanding of morphological plant *sex*.⁸⁰ Lloyd's redefinitions of male and female in relation to the former mean the identification of a plant's gender may well conflict with the identification of its sex, traditionally understood: 'It is frequently assumed that plants in one population and with the same general sex form are identical in their gender, but different plants of the same general phenotype can vary considerably in their precise gender.'⁸¹ But if Lloyd's position is clearly that functional gender is superior to phenotypic sex as a measure of sexual behaviour in plants,⁸² what is the status of the idea of morphological plant sex in his work? Or what can we now say about that idea?

Lloyd does not reject the idea of morphological plant sex entirely but sees it as an indicator of last resort when the information that would enable one to calculate plant gender is unavailable. Given the complexity and time investment required to calculate functional gender, Lloyd accepts that designations of morphological plant sex will sometimes still be used and indeed are useful, just as the 'typological' distinction between sexual systems, although crude, is useful:

Gynodioecy is not a uniform breeding system but a wide range of conditions... [it] nevertheless remains a useful term to describe a class of plants with distinctive properties in common... neither the presence of a continuum in sex functions nor the use of quantitative measures to describe these functions removes the need for a simple verbal description of such populations.⁸³

The characterization of the morphological sex categories as 'verbal description', 'verbal label[s]' or 'verbal terms'⁸⁴ contrasts them with the theoretically determined *concept* of functional plant gender and of the *concepts* of male and female associated with it. That is, whereas plant 'gender' and Lloyd's redefinitions of male and female *for plants* are basic *concepts* grounding his investigation of plant sexual behaviour, the terms of morphological plant sex are just *words* applied to plants for convenience's sake. Moreover, they are words derived from an animal model. Thus Bailey's plea for the continued use of the old-time untechnical vocabulary is respected, but it is clearer what this use involves: a pragmatic compromise which is very far from the 'literal' depiction of plant sex hailed by the traditional historians of botany, being, on the contrary, 'a manner of speaking' about plants borrowed from another form of life.

Lloyd's concept of plant gender, on the other hand, and the definitions of male and female for plants that follow from it, attempts to move beyond the animal model to propose specifically vegetal ways of thinking about plant sexuality. Lloyd, of course, is only concerned with the consequences of this reconceptualization for the plant sciences, and particularly for a more adequate approach to the understanding of sexual reproductive strategies in plant populations. But the work can also legitimately be interpreted as a series of propositions for plant philosophy concerning vegetal sex, propositions that also have implications for the way that we understand sex more generally (as we will discuss further in the Epilogue).

First, what does Lloyd's reconceptualization of plant gender allow us to *know* about vegetal sex? Lloyd utilizes the traditional categories of the animal model: male and female. But the redefinition of these categories in functional, quantitative terms breaks the presumption of the animal model which, we can now see, operates according to a kind of ontic *emboîtement*: the gamete is the smallest male or female unit, nesting (at least at first) within the larger male or female unit of the organ, which is itself nested in or part of the largest male or female unit, the individual. In the animal model the direction of identification moves from the gamete to the organ to the individual, or the determining criterion of the identification of the largest male and female unit is based in its ontological relation to the smallest (the being of the smallest determines the being of the largest). Nevertheless, the logic of the individual prevails at each level of the ontological determination of sex: a gamete, an organ, an individual male or female animal. The animal model is fundamentally a system of ontological metonymy controlled by the logical priority of individuality.

Lloyd's concept of plant gender, though it has its biological basis in the idea of the gamete, is, on the other hand, a concept that is conceived in terms of multiplicity. The direction of identification moves from the population (a dynamic multiplicity) and the potential distribution of the multiplicity of gametes within it, to assign a gender *value* to an individual. As a value, plant gender is not an attribute of a plant subject but its coordinates, its current gender position within a population. This plant model is fundamentally a system of mathematical coordination controlled by the logical priority of multiplicity (but not collectivity). It is not an ontology of plant sex; there is no ontology of plant sex. It is the mobilization of a relational epistemology where what is known is not the literal truth of a plant's sex but the gender value of its current allocation in relation to a population.

Lloyd is clear that a gender value is an estimate and that the identification of the 'effective' (i.e. actual) gender of a plant may be impossible. But, philosophically, this conception of plant gender is not proposed *faute de mieux*. It is not a temporary stand-in while we wait for the truth of plant sex to be delivered up in the form of an ontology that we can recognize. It is more fundamentally an acknowledgement of the alterity of plant life and the alterity of plant sexuality. Lloyd's formulas for the measurement of plant gender yield estimates that acknowledge, in effect, that we do not know the 'measure' of plant sex because we have not yet rid ourselves of the conceptual instruments attuned to the key of animal being.

But are these instruments out of tune, even for us?